FUNDAMENTALS OF PROGRAMMING & DATA MANAGEMENT MOD. B

Course Presentation

by

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DIETI UNIVERSITA' DEGLI STUDI DI NAPOLI FEDERICO II

DIPARTIMENTO DI INGEGNERIA ELETTRICA E TECNOLOGIE DELL'INFORMAZIONE

Teaching Materials

- Textbooks:
  1. Chianese, Moscato, Picariello, Sansone. "Database systems and applications." Apogeo Education-Maggioli Editore. September 2015.
  2. Database Systems: Design, Implementation, & Management di Carlos Coronel (Autore) , Steven Morris (Autore) Cengage Learning, Inc, 2022.
  3. Avi Silberschatz, Henry F. Korth, S. SudarshanDatabase System Concepts,Seventh Edition (https://db-book.com/slides-dir/)
- Course Slides
- Lecture notes

AI and Big Data – Course Syllabus

- Instructor:
  - **Prof. Vincenzo Moscato**
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  - **10:30-12:30 Tuesday**
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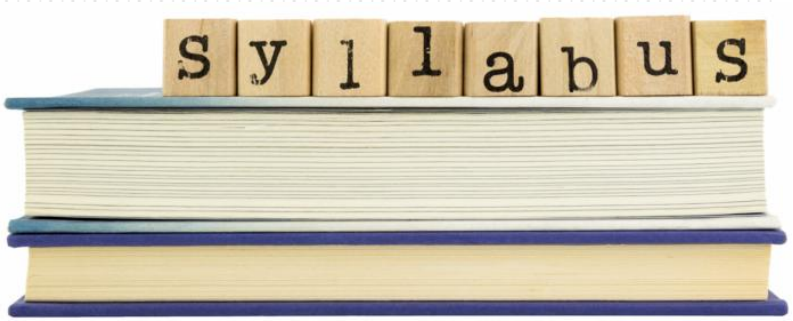

Syllabus: Topics

- Information systems. Information and computer systems.
- Databases and management systems (DBMS).
- The relational model. Relationships and tables. Databases and integrity constraints. Definition of data in SQL.
- The Entity Relationship model. Database design. Entities, associations and attributes. Conceptual design and examples. From conceptual schema to relational schema. Schema review. Translation into the logical model. The Advanced Relational Entity Model. Inheritance: superclasses and subclasses. Hierarchies of generalization and specialization. Resolution of hierarchies.
- Operations. Setwise operations. Changing the state of the database. Operations Relational in procedural and declarative form (SQL). Selection, Projection, Join. Renaming and use of variables. Aggregation and grouping functions. Queries Set and nested. Views. SQL query syntax. The complete syntax of Insert, Update and Delete.
- Normal Forms. Redundancies and anomalies in editing a relation. Functional Dependencies. Constraints and functional dependencies; complete dependencies. The three normal forms and techniques of decomposition. The normal form of Boyce and Codd.
- SQL and programming languages. ODBC, JDBC, triggers.
- DBMS technologies and transaction management.
- Data Analytics.

Practical Lectures

- Database Design
- SQL and PL/SQL programming by Oracle Live Cloud DBMS
- Microsoft Power BI
- Text_to_SQL (via LLM)
- Homeworks and Final Project
  - Supported by my PhD students and collaborators
    - Antonio Laudante, Francesco Di Serio

Final Examination

- Oral Interview on Course Topics
- Homeworks:
  - Database Design: from natural language data specification to relational database creation
  - Data Analysis by querying database
  - Dashboard Implementation using Power BI
- Final project (Teams of 3 students)
- Thesis Request Available for students